LAB SHEET – 10

CampusConnect – Secure Full-Stack Application

Task 1 – Authentication & RBAC

```
const UserSchema = new mongoose.Schema({
 name:String, email:{type:String,unique:true},
 passwordHash:String, role:{type:String,enum:["ADMIN","STUDENT"]}
});

function auth(req,res,next){
 const t=req.headers.authorization?.split(" ")[1];
 try{req.user=jwt.verify(t,process.env.JWT_SECRET);next();}
 catch(e){res.status(401).json({message:"Unauthorized"});}
}
const admin=(req,res,next)=>req.user.role==="ADMIN"
 ? next():res.status(403).json({message:"Forbidden"});
const hash=await bcrypt.hash(password,10);
const token=jwt.sign({id:user._id,role:user.role},
 process.env.JWT_SECRET,{expiresIn:"15m"});
```

Task 2 – Real-Time Notifications

```
const io=new Server(server,{cors:{origin:process.env.FRONTEND_URL}});
io.use((s,next)=>{try{s.user=jwt.verify(s.handshake.auth.token,SECRET);next()}
catch(e){next(new Error("Unauthorized"));}});
io.on("connection",s=>s.user.role==="STUDENT"&&s.join("students"));
io.to("students").emit("new-announcement",announcement);
const socket=io(API,{auth:{token}});
socket.on("new-announcement",a=>alert(a.title));
```

Task 3 – Redis Caching

```
const redis=createClient({url:process.env.REDIS_URL});
await redis.connect();
app.get("/api/events",async(req,res)=>{
 const c=await redis.get("events");
 if(c)return res.json(JSON.parse(c));
 const events=await Event.find();
 await redis.setEx("events",60,JSON.stringify(events));
 res.json(events);
});
await redis.del("events");
```

Task 4 – React + Redux

```
const store=configureStore({
 reducer:{auth:authReducer,events:eventReducer}
});
api.interceptors.request.use(c=>{
 c.headers.Authorization=`Bearer ${token}`; return c;
});
function PrivateRoute({children,role}){
 const u=useSelector(s=>s.auth);
 if(!u.token)return <Navigate to="/login"/>;
 return role&&u.role!==role?<Navigate to="/dashboard"/>:children;
```

```
}
events.filter(e=>e.title.toLowerCase().includes(search.toLowerCase()));
```

Task 5 – Security Hardening

```
app.use(helmet());
app.use(cors({origin:process.env.FRONTEND_URL}));
app.use(express.json());
app.use("/api/auth/login",rateLimit({
 windowMs:15*60*1000,max:5
}));
const schema=z.object({
 email:z.string().email(), password:z.string().min(8)
});
const result=schema.safeParse(req.body);
if(!result.success)
 return res.status(400).json({errors:result.error.issues});
```

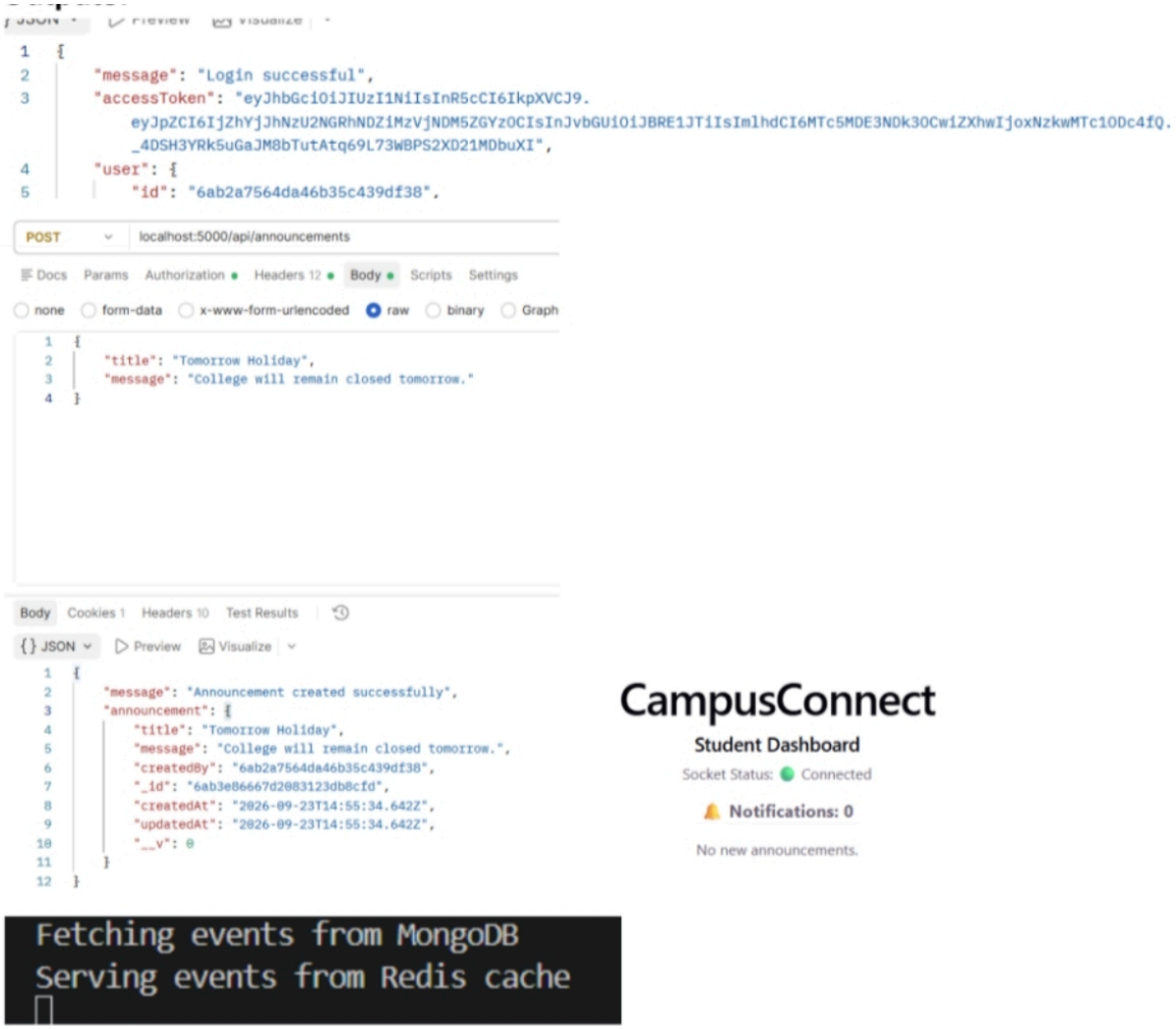
Task 6 – Testing

```
test("wrong password",async()=>{
 const r=await request(app).post("/api/auth/login")
  .send({email:"a@test.com",password:"wrong"});
 expect(r.status).toBe(401);
});
test("no token",async()=>{
 const r=await request(app).get("/api/events/admin");
 expect(r.status).toBe(401);
});
test("student forbidden",async()=>{
 const r=await request(app).post("/api/events")
  .set("Authorization","Bearer STUDENT_TOKEN");
 expect(r.status).toBe(403);
});
test("login fields",()=>{
 render(<Login/>);
 expect(screen.getByPlaceholderText(/email/i)).toBeInTheDocument();
});
```

Task 7 – Docker & Deployment

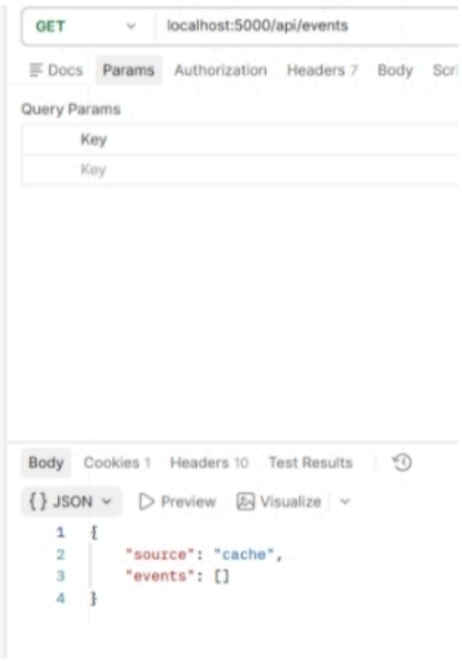
```
FROM node:20-alpine
WORKDIR /app
COPY package*.json ./
RUN npm ci --omit=dev
COPY . .
CMD ["node","server.js"]
FROM node:20-alpine AS build
WORKDIR /app
COPY . .
RUN npm install && npm run build
FROM nginx:alpine
COPY --from=build /app/dist /usr/share/nginx/html
services:
 mongodb:
  image:mongo:7
  volumes:[mongo_data:/data/db]
 redis:
```

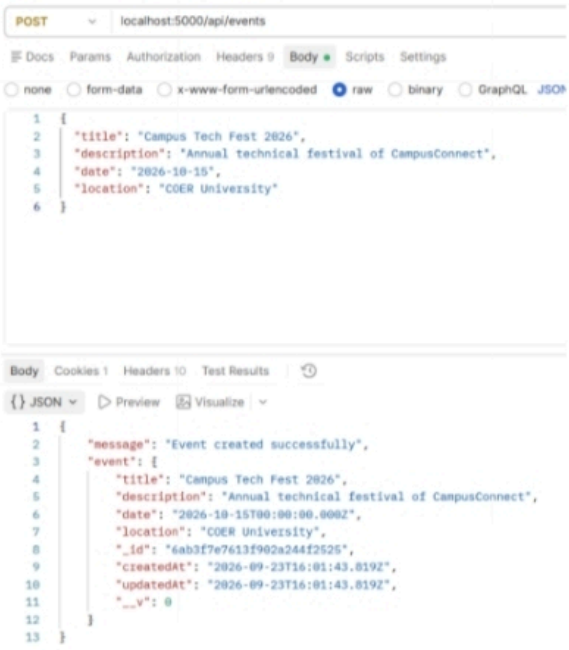
```
  image:redis:7-alpine
 backend:
  build:./backend
  depends_on:[mongodb,redis]
 frontend:
  build:./frontend
  ports:["80:80"]
volumes:
 mongo_data:
```

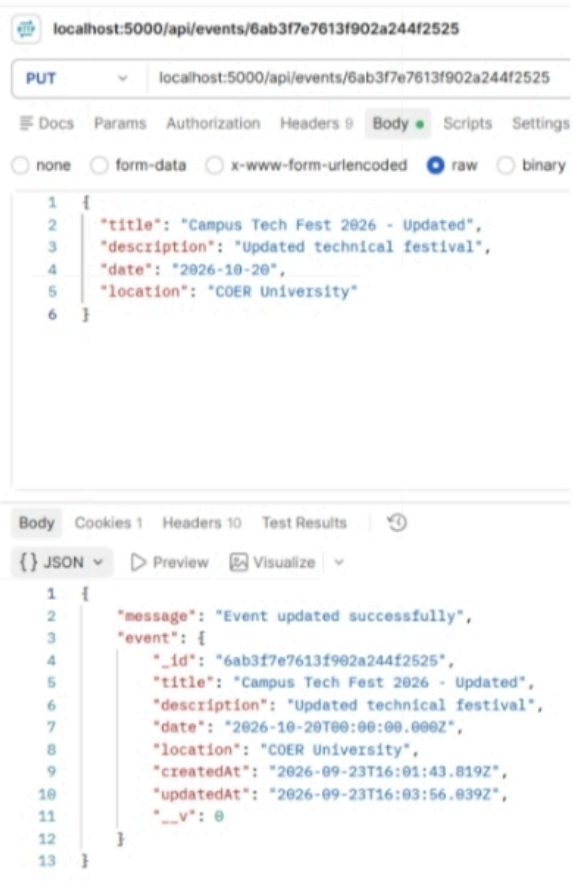

{
"message": "Login successful",
"accessToken": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.
eyJpZCI6IjZhYjJhNzU2NGRhNDZiMzVjNDM5ZGYzOCIsInJvbGUiOiJBRE1JTiIsImlhdCI6MTc5MDE3NDk3OCwiZXhwIjoxNzkwMTc1ODc4fQ.
_4DSH3YRk5uGaJM8bTutAtq69L73WBPS2XD21MDbuXI",
"user": {
"id": "6ab2a7564da46b35c439df38",
POST
localhost:5000/api/announcements
Docs
Params
Authorization
Headers 12
Body
Scripts
Settings
none
form-data
x-www-form-urlencoded
raw
binary
Graph
{
"title": "Tomorrow Holiday",
"message": "College will remain closed tomorrow."
}
Body
Cookies 1
Headers 10
Test Results
{} JSON
Preview
Visualize
{
"message": "Announcement created successfully",
"announcement": {
"title": "Tomorrow Holiday",
"message": "College will remain closed tomorrow.",
"createdBy": "6ab2a7564da46b35c439df38",
"_id": "6ab3e86667d2083123db8cfd",
"createdAt": "2026-09-23T14:55:34.642Z",
"updatedAt": "2026-09-23T14:55:34.642Z",
"__v": 0
}
}

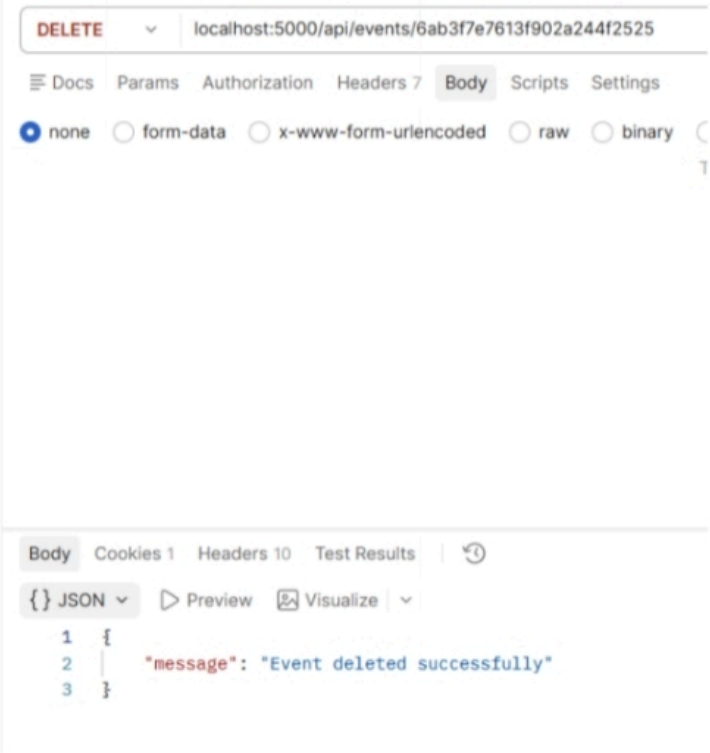
Fetching events from MongoDB
Serving events from Redis cache

CampusConnect
Student Dashboard
Socket Status: Connected
Notifications: 0
No new announcements.


GET
localhost:5000/api/events
Docs
Params
Authorization
Headers 7
Body
Query Params
Key
Key
Body
Cookies 1
Headers 10
Test Results
JSON
Preview
Visualize
{
"source": "cache",
"events": []
}


POST
localhost:5000/api/events
Docs
Params
Authorization
Headers 9
Body
Scripts
Settings
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
{
"title": "Campus Tech Fest 2026",
"description": "Annual technical festival of CampusConnect",
"date": "2026-10-15",
"location": "COER University"
}
Body
Cookies 1
Headers 10
Test Results
JSON
Preview
Visualize
{
"message": "Event created successfully",
"event": {
"title": "Campus Tech Fest 2026",
"description": "Annual technical festival of CampusConnect",
"date": "2026-10-15T00:00:00.000Z",
"location": "COER University",
"_id": "6ab3f7e7613f902a244f2525",
"createdAt": "2026-09-23T16:01:43.819Z",
"updatedAt": "2026-09-23T16:01:43.819Z",
"__v": 0
}
}


localhost:5000/api/events/6ab3f7e7613f902a244f2525
PUT
localhost:5000/api/events/6ab3f7e7613f902a244f2525
Docs
Params
Authorization
Headers 9
Body
Scripts
Settings
none
form-data
x-www-form-urlencoded
raw
binary
{
"title": "Campus Tech Fest 2026 - Updated",
"description": "Updated technical festival",
"date": "2026-10-20",
"location": "COER University"
}
Body
Cookies 1
Headers 10
Test Results
JSON
Preview
Visualize
{
"message": "Event updated successfully",
"event": {
"_id": "6ab3f7e7613f902a244f2525",
"title": "Campus Tech Fest 2026 - Updated",
"description": "Updated technical festival",
"date": "2026-10-20T00:00:00.000Z",
"location": "COER University",
"createdAt": "2026-09-23T16:01:43.819Z",
"updatedAt": "2026-09-23T16:03:56.039Z",
"__v": 0
}
}


DELETE
localhost:5000/api/events/6ab3f7e7613f902a244f2525
Docs
Params
Authorization
Headers 7
Body
Scripts
Settings
none
form-data
x-www-form-urlencoded
raw
binary
Body
Cookies 1
Headers 10
Test Results
JSON
Preview
Visualize
{
"message": "Event deleted successfully"
}


CampusConnect
Login
Email
Password
Login